

Testing the combined effects of berberine and cyanidin-3-glucoside chloride on insulin regulation in Drosophila melanogaster

Introduction

Problem: Type 2 Diabetes Mellitus (T2DM) affects over 483 million people globally. Biologically, it is mainly due to insulin resistance which is the process in which the body's cells become less sensitive to insulin. In turn, glucose is insufficiently regulated. Over time, it causes hyperglycemic conditions, oxidative stress, and damaged beta cells. Even though there is no cure, some prescription drugs provide alleviation treatment, but they often have a high cost and many side effects.

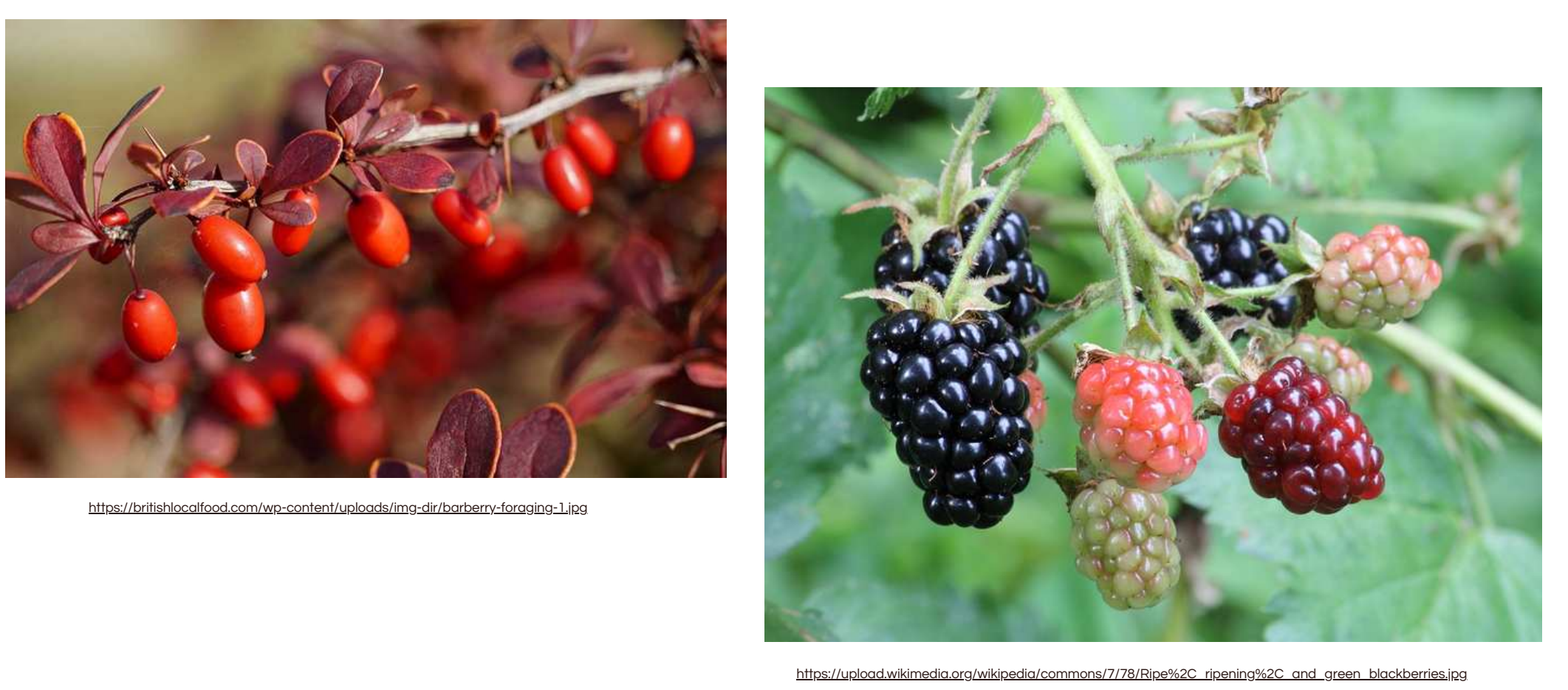
Compounds: Bioactive compounds have been found to have the potential to do this. These are natural substances that have a positive effect on a living organism. Many bioactive compounds have been identified that can improve glucose homeostasis and insulin signaling. A 2025 study conducted by Ye et al. found these characteristics within cyanidin-3-glucoside chloride. Similarly, a 2024 study conducted by Arjun et al. found the same with berberine. The problem is that the effects aren't strong enough for clinical application or human testing.

Distinct Mechanism: This reveals the possibility of combining bioactive compounds for a stronger result, but it remains unexplored. Foundational studies have shown that different bioactive compounds including alkaloids and flavonoids have distinct metabolic mechanisms during insulin secretion. C3G enhances endocrine signals upstream of insulin releases. On the other hand, berberine may improve downstream insulin pathway sensitivity and metabolic response. This difference suggests they would have a more powerful effect on reversing insulin resistance if combined.

Research Question: Can berberine and cyanidin 3-glucoside chloride regulate insulin expression in Drosophila melanogaster in a high-sugar diet?

Hypothesis

If Drosophila melanogaster are fed a high sugar diet, then the combination of cyanidin-3-glucoside chloride and berberine will improve insulin resistance more than the individual compounds will measured by the the difference in dilp2 levels and glucose levels compared between experimental groups and standard diet because cyanidin-3-glucoside chloride enhances upstream insulin signaling, while berberine enhances downstream insulin release.



Method

- 01 Toxicity Assay**
 - Find the Maximum Non-Toxic Concentration for BBR and C3G
- 02 Set up diets**
 - Standard Diet (SD): Corn syrup, yeast, flour, cornmeal, agar
 - High Sugar Diet (HSD): SD + 1M sucrose
- 03 Fly Setup**
 - Maintain diet for 20 days (10 days HSD, 10 days treatment)
- 04 qPCR**
 - Collect heads, extract RNA, reverse transcription into cDNA, conduct qPCR
 - Primers: dilp2 and rp49
- 05 Analyze Results**
 - Calculate relative expression from Ct
 - Compare expression levels to draw conclusions

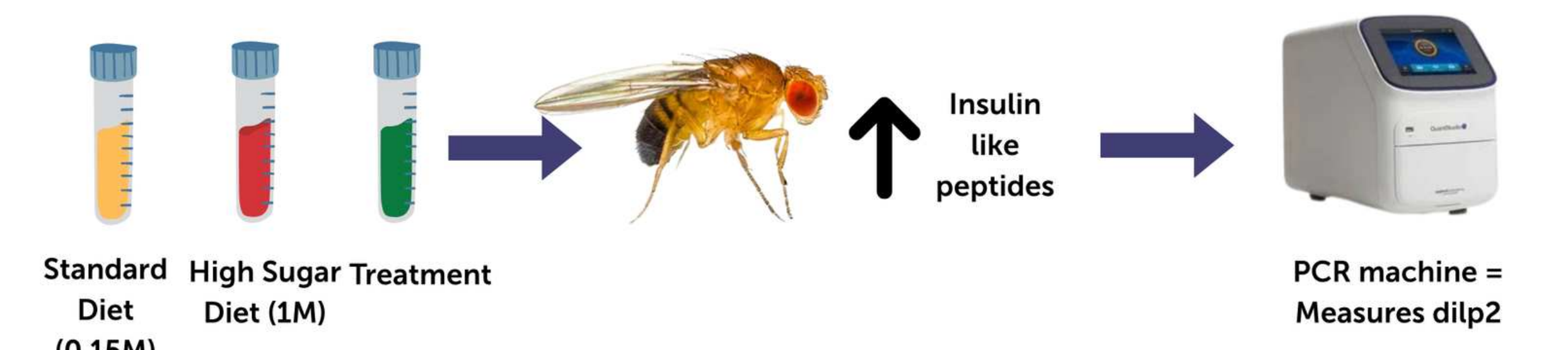


Figure 1: Visualization of Method

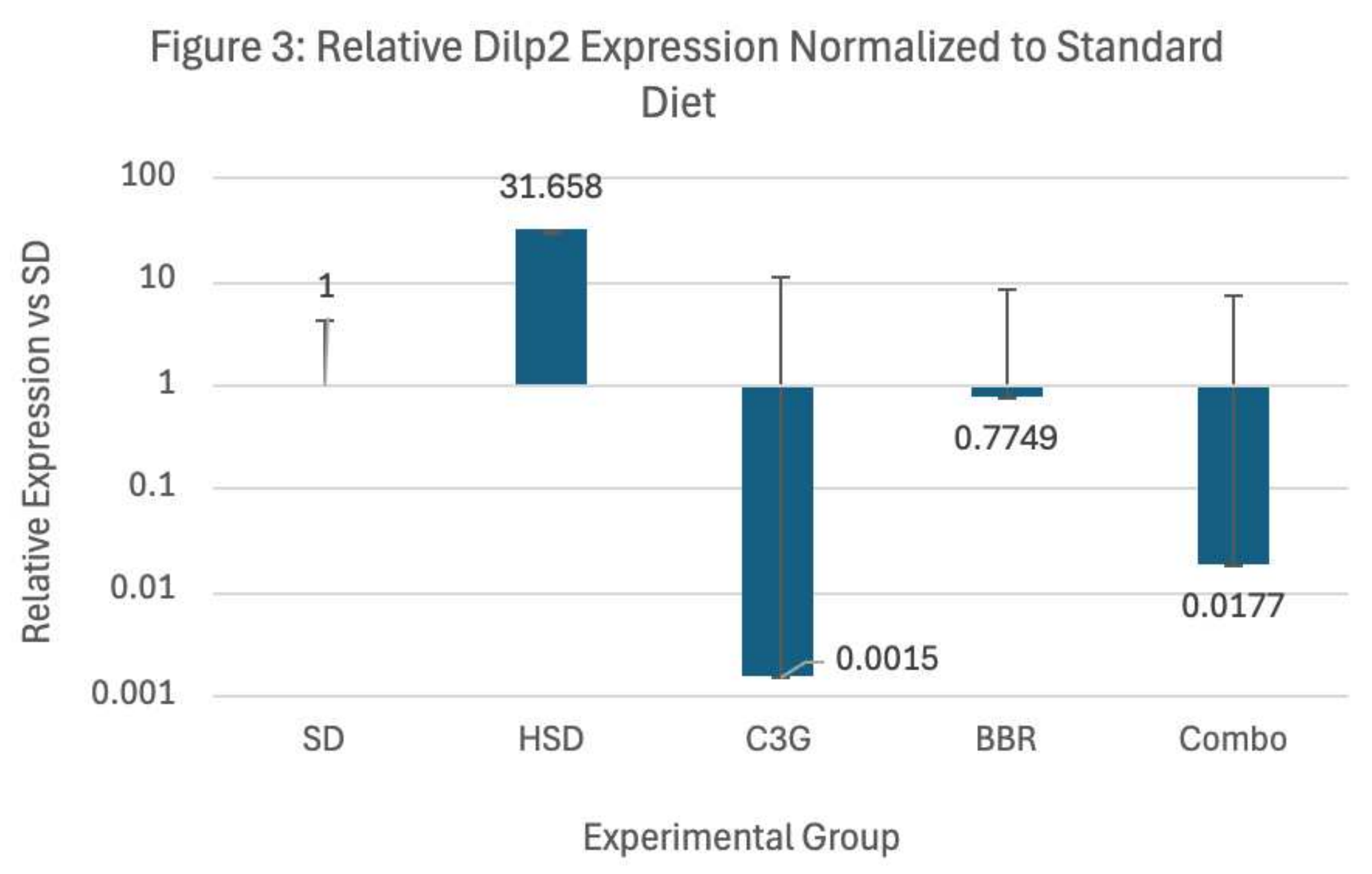
- Variables**
- Independent Variable: Diet
 - Dependent Variable: Dilp2 Expression
- Controls**
- Positive Control: High Sugar Diet
 - Negative Control: Standard Diet



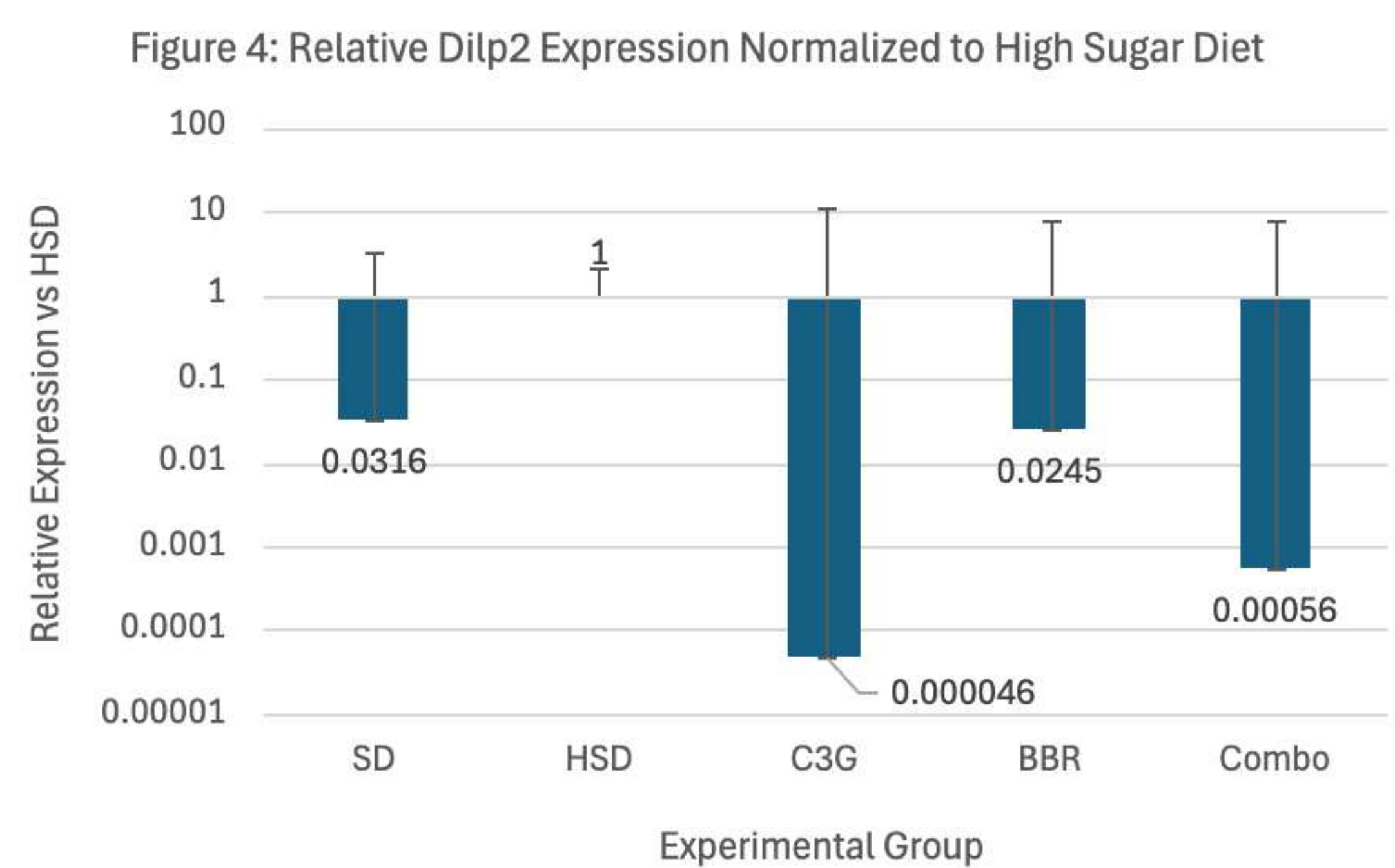
Figure 2: Visualization of Fly Food Creation and Expansion

Data/Results

Table 1: Qubit Assay for RNA Extraction		Table 2: Relative Expressions of Dilp2 Using qPCR					
Groups	Amount of RNA(ng/uL)	Experimental Group	Mean ΔCt	ΔΔCt vs SD	Relative Expression vs SD	ΔΔCt vs HSD	Relative Expression vs HSD
SD	4.92	Standard Diet	0.1020	0.0000	1.0000	4.9845	0.0316
HSD	15.60	High Sugar Diet	-4.8825	-4.9845	31.6580	0.0000	1.0000
C3G	7.60	Cyanidin-3-Glucoside Chloride	9.5117	9.4097	0.0015	14.3942	0.000046
BBR	3.31	Berberine	0.4700	0.3680	0.7749	5.3525	0.0245
Combo 1	14.50	Combination	5.9218	5.8198	0.0177	10.8043	0.00056
Combo 2	8.76						



A logarithmic scale was used because relative expression values varied widely between experimental groups, so the scale allows all the expression levels to be displayed on the graph. The bars represent fold changes when compared to the Standard Diet in Figure 3 or High Sugar Diet in Figure 4.



Conclusions

The qPCR results partially support the original hypothesis that the combined treatment of berberine and cyanidin-3-glucoside chloride would improve insulin resistance more than the individual treatments. First, the high sugar diet shows a large increase in relative dilp2 expression from Table 3 and Figure 4. It increases by about 31x in the high sugar diet from the standard diet. This shows that this diet created metabolic stress and hyperglycemic conditions to where it altered the insulin related gene function. Because dilp2 is an insulin-like peptide, this increase may reflect a compensatory response to excess dietary sugar.

Compared to the high sugar diet, all the treatment groups show lower levels of dilp2 expressions. This suggests that the treatments helped reduce the insulin regulation disruption caused by the high sugar diet. Analyzing Figure 4, berberine restores insulin levels closest to the standard diet levels with a relative expression level of 0.7749. However, the combined treatment appears to show a stronger reduction, with a relative expression of 0.0177, than berberine alone and a weaker reduction than cyanidin-3-glucoside chloride alone (relative expression of 0.0015). This supports the idea that the combined treatment had a moderating effect rather than a synergistic effect. Overall, this provides partial support for the hypothesis because the combined treatment provides a different effect than both compounds individually. However, the synergism needs to be confirmed through more trials and an additional glucose assay.

However, the results do not fully show that the combined treatment reversed insulin resistance. Future steps that need to be conducted include a paired glucose assay, more biological replicates, and more technical replicates. Many of the technical replicates showed high Ct variability, so there could be large error associated with these replicates.

References

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